DATE:24/01/2026

EX-01

Windows Fundamental-1

AIM:

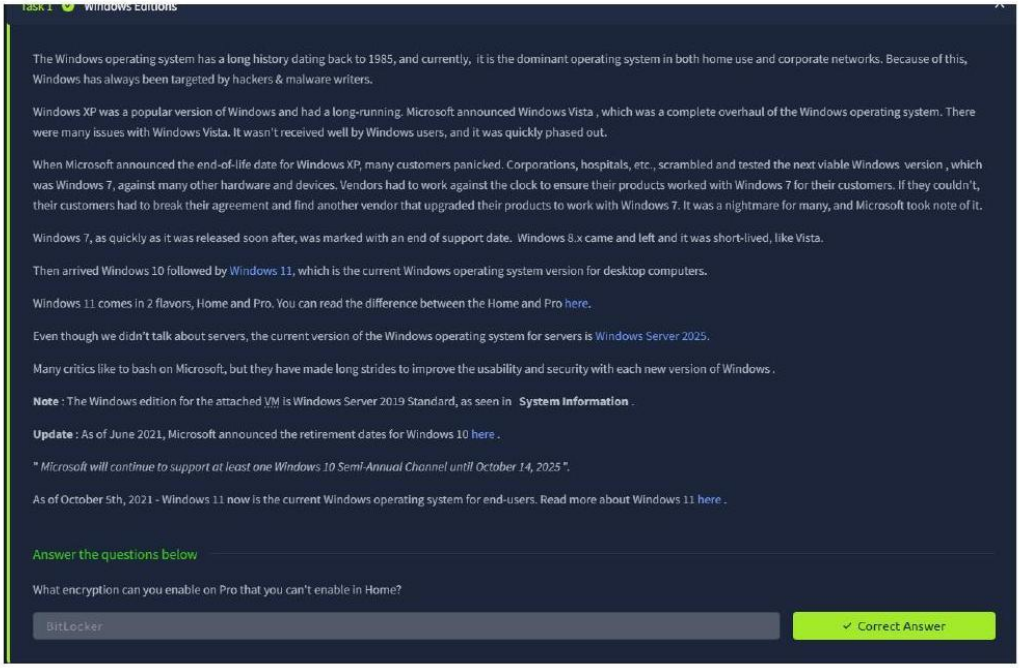
To gain hands-on familiarity with the **core components of the Windows operating system**, including the desktop environment, NTFS file system architecture, user account management, and essential system utilities such as Control Panel and Task Manager.

PROCEDURE:

- Launch the virtual machine in the TryHackMe lab environment.
- Establish a Remote Desktop connection to the Windows system.
- Navigate and explore the Windows Desktop, NTFS File System, and File Explorer.
- Access the Control Panel and Settings to examine system configurations.
- Investigate User Account Control (UAC) and interact with system tools including Task Manager.

TASK-1:

TASK-1:



Task 1 Windows Editions

The Windows operating system has a long history dating back to 1985, and currently, it is the dominant operating system in both home use and corporate networks. Because of this, Windows has always been targeted by hackers & malware writers.

Windows XP was a popular version of Windows and had a long-running. Microsoft announced Windows Vista , which was a complete overhaul of the Windows operating system. There were many issues with Windows Vista. It wasn't received well by Windows users, and it was quickly phased out.

When Microsoft announced the end-of-life date for Windows XP, many customers panicked. Corporations, hospitals, etc., scrambled and tested the next viable Windows version , which was Windows 7, against many other hardware and devices. Vendors had to work against the clock to ensure their products worked with Windows 7 for their customers. If they couldn't, their customers had to break their agreement and find another vendor that upgraded their products to work with Windows 7. It was a nightmare for many, and Microsoft took note of it.

Windows 7, as quickly as it was released soon after, was marked with an end of support date. Windows 8.x came and left and it was short-lived, like Vista.

Then arrived Windows 10 followed by Windows 11, which is the current Windows operating system version for desktop computers.

Windows 11 comes in 2 flavors, Home and Pro. You can read the difference between the Home and Pro here.

Even though we didn't talk about servers, the current version of the Windows operating system for servers is Windows Server 2025.

Many critics like to bash on Microsoft, but they have made long strides to improve the usability and security with each new version of Windows .

Note : The Windows edition for the attached VM is Windows Server 2019 Standard, as seen in **System Information** .

Update : As of June 2021, Microsoft announced the retirement dates for Windows 10 here .

" Microsoft will continue to support at least one Windows 10 Semi-Annual Channel until October 14, 2025 ".

As of October 5th, 2021 - Windows 11 now is the current Windows operating system for end-users. Read more about Windows 11 here .

Answer the questions below

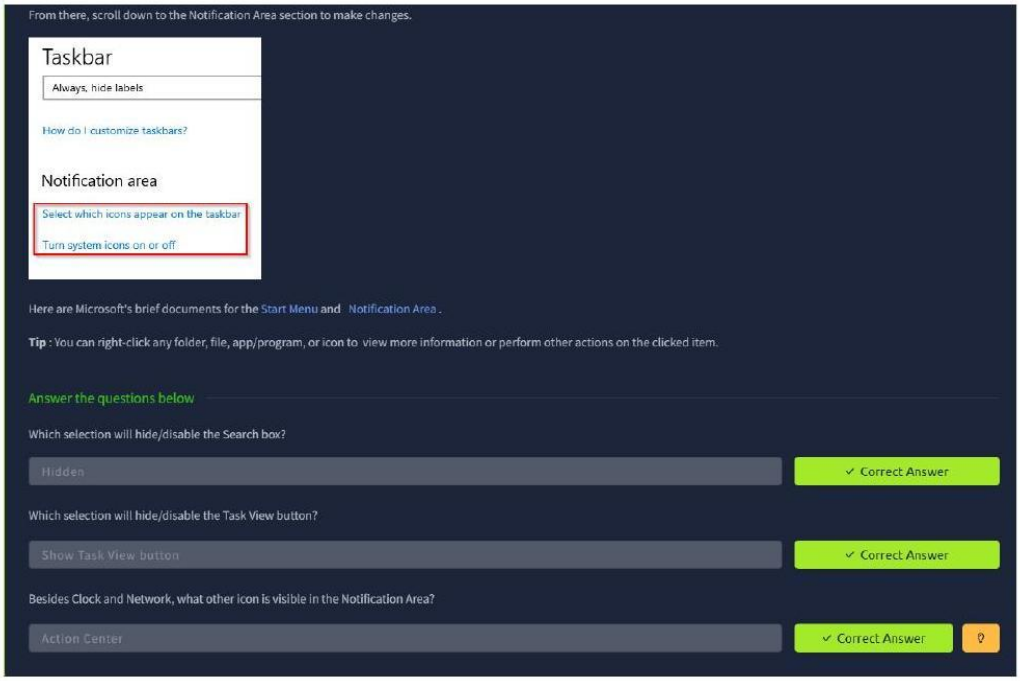
What encryption can you enable on Pro that you can't enable in Home?

BitLocker

✓ Correct Answer

TASK-2:

TASK-2:

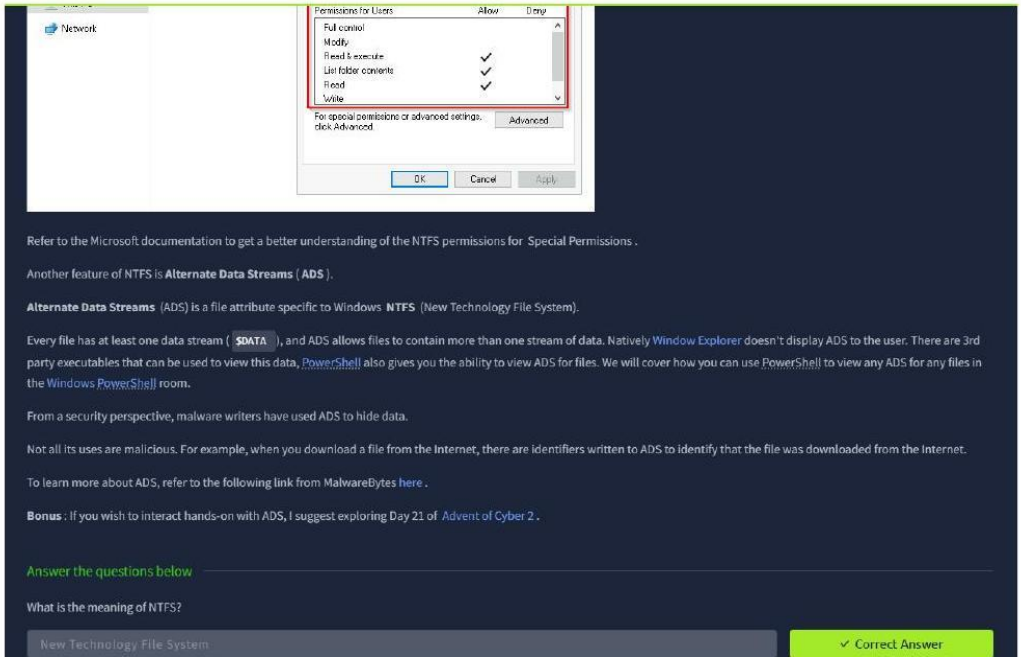

From there, scroll down to the Notification Area section to make changes.
Taskbar
Always, hide labels
How do I customize taskbars?
Notification area
Select which icons appear on the taskbar
Turn system icons on or off
Here are Microsoft's brief documents for the Start Menu and Notification Area .
Tip : You can right-click any folder, file, app/program, or icon to view more information or perform other actions on the clicked item.
Answer the questions below
Which selection will hide/disable the Search box?
Hidden
✓ Correct Answer
Which selection will hide/disable the Task View button?
Show Task View button
✓ Correct Answer
Besides Clock and Network, what other icon is visible in the Notification Area?
Action Center
✓ Correct Answer

TASK-3:

User : administrator
Password : letmein123!
Remote Desktop Preference
Profile
Name
Quick Connect
Group
Protocol
RDP - Remote Desktop Protocol
Pre Command
Post Command
/path/to/command -opt1 arg %h %u %t -opt2 %U %p %g
Basic
Advanced
SSH Tunnel
Server
10.10.90.149
User name
administrator
User password
Domain
Resolution
Use client resolution
Custom
Color depth
RemoteFX (32 bpp)
Accept the Certificate when prompted, and you should be logged into the remote system now.
Note : The virtual machine may take up to 3 minutes to load.
Answer the questions below
Read above and start the virtual machine.
No answer needed
✓ Correct Answer

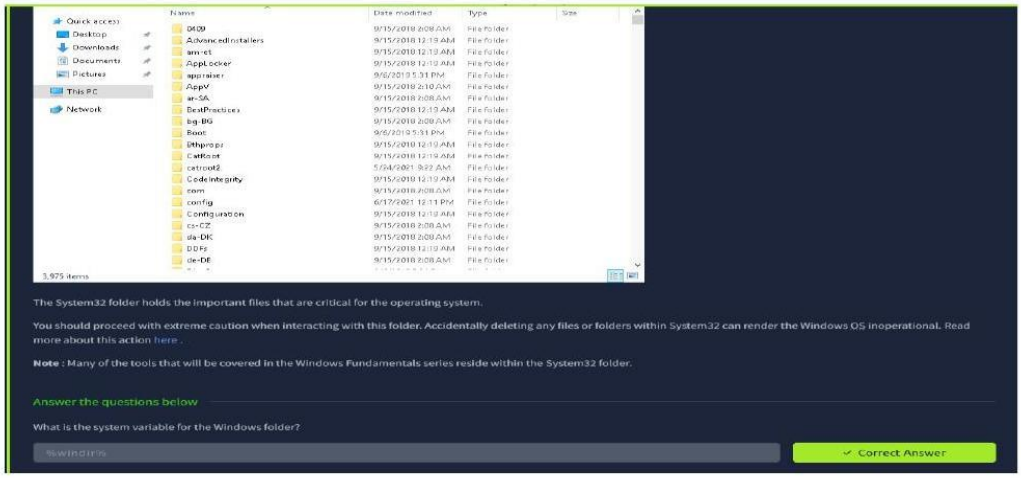
TASK-4:

TASK-4:



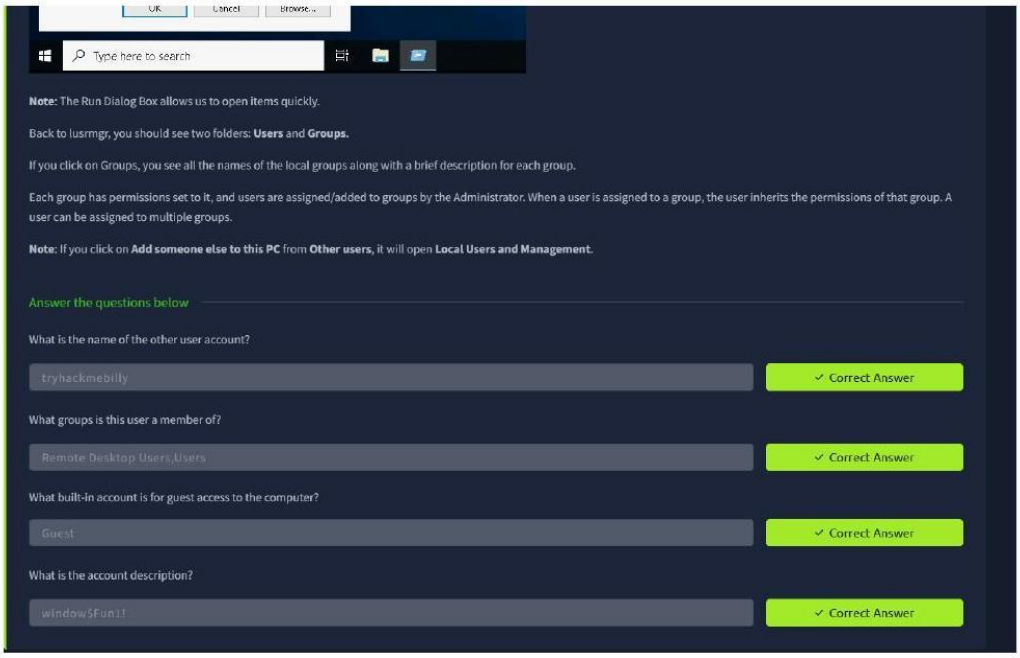
TASK-5:

TASK-5:



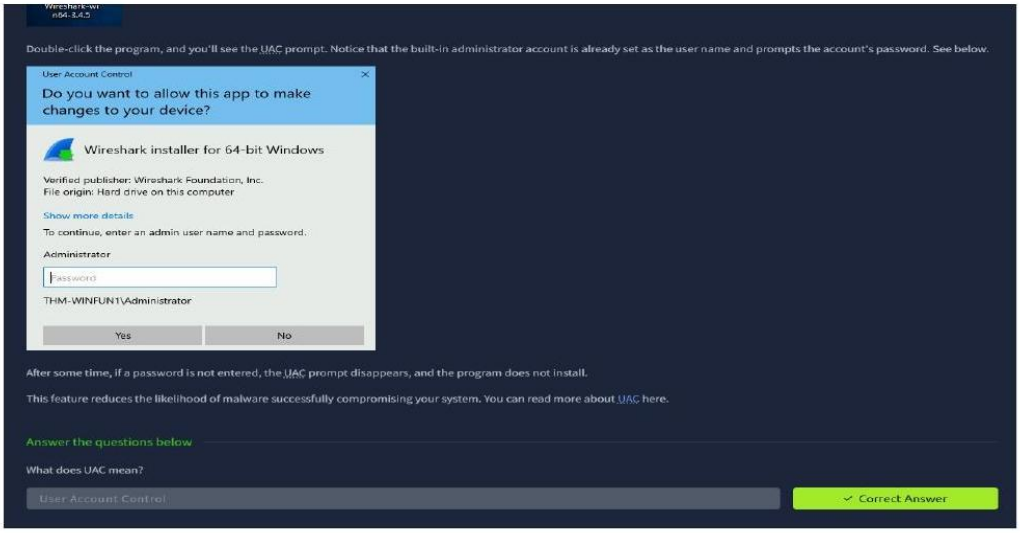
TASK-6:

TASK-6:



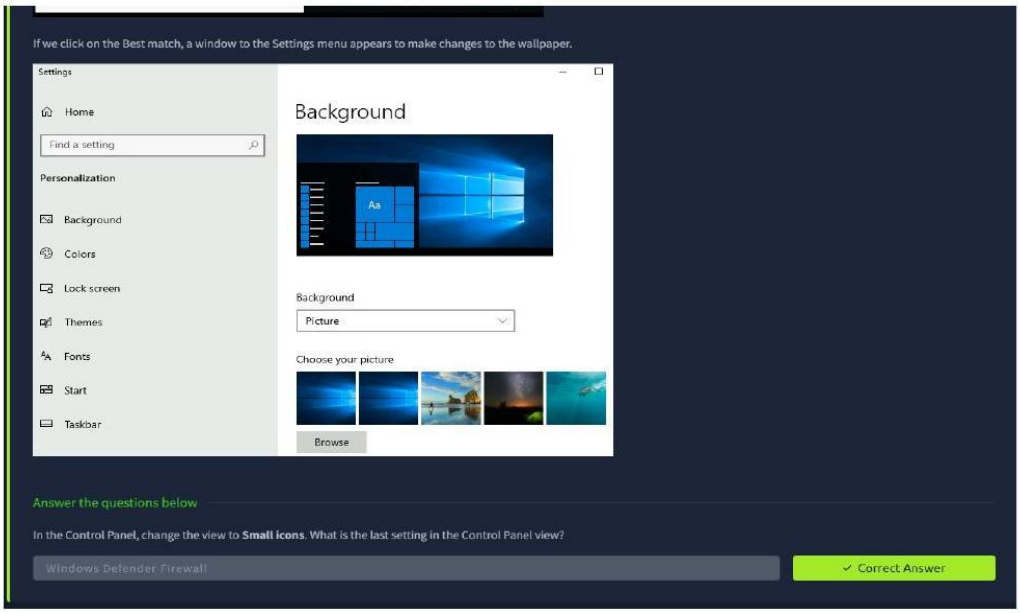
TASK-7:

TASK-7:



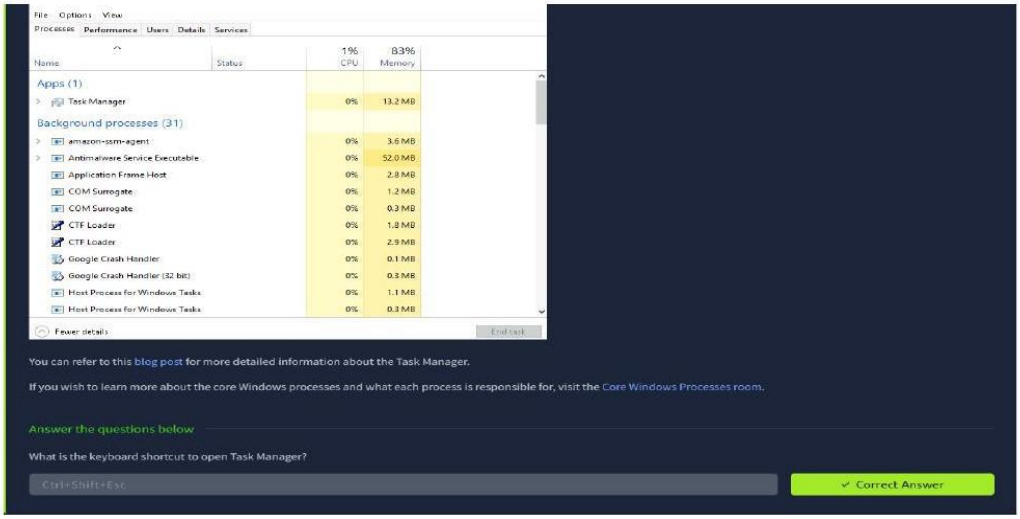
TASK-8:

TASK-8:



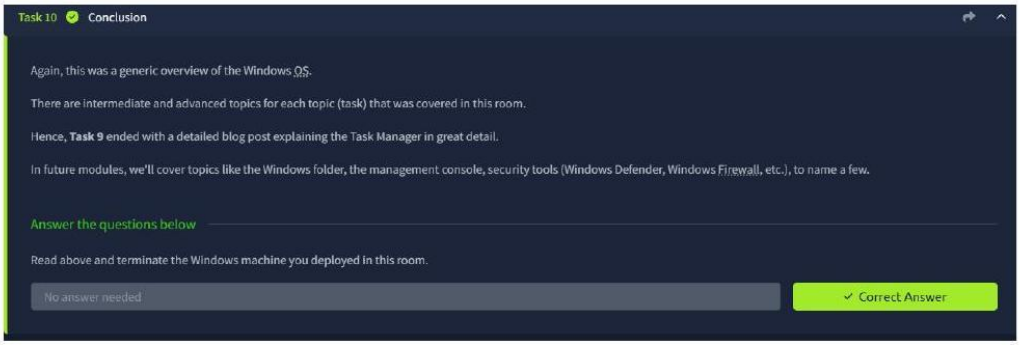
TASK-9:

TASK-9:



TASK-10:

TASK-10:



RESULT:

Successfully explored and understood the fundamental components of the Windows operating system, covering the desktop interface, NTFS file system, user accounts, and system tools through Task Manager.

Windows Fundamentals-2

AIM:

To develop a practical understanding of **Windows administrative utilities and system management tools**, including System Configuration, Computer Management, Resource Monitor, and the Windows Registry, for effective system administration.

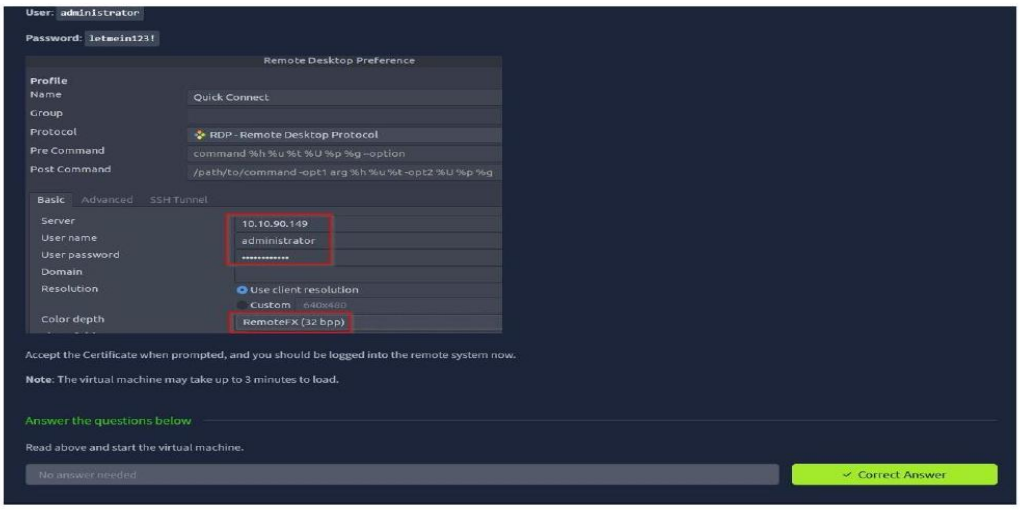
PROCEDURE:

- Start and initialize the designated lab virtual machine.
- Log into the system using the provided lab credentials.
- Open and examine System Configuration and Computer Management tools.
- Explore built-in utilities such as Task Scheduler and Event Viewer.
- Use Resource Monitor to analyze real-time CPU, disk, memory, and network usage.
- Review system logs and active processes, then complete the lab questions.

TASK-1:

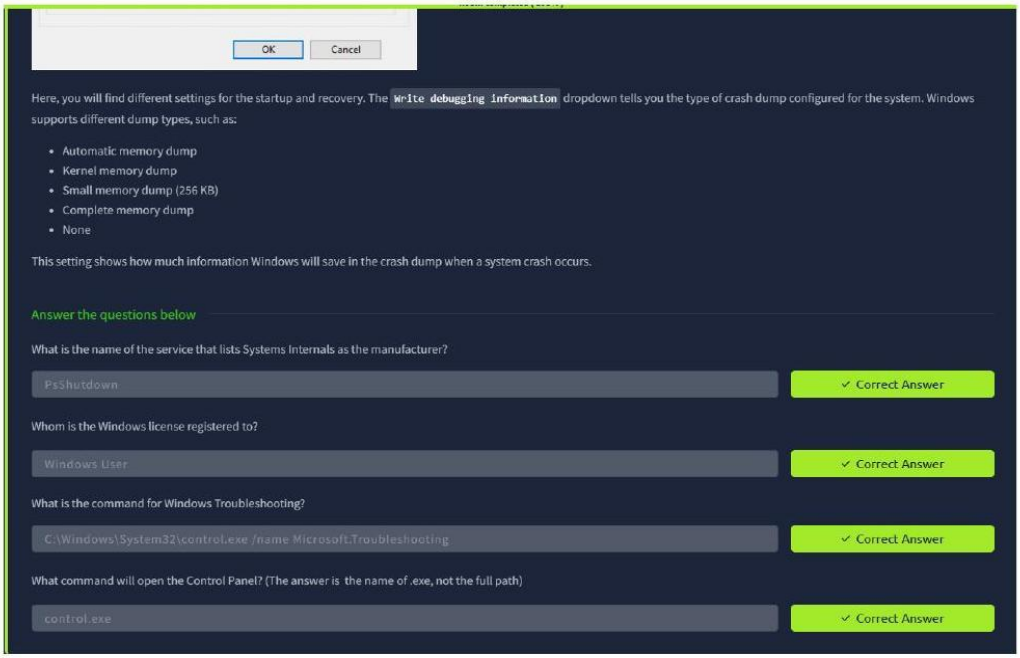
- Connect to the system using the provided credentials.
- Open **System Configuration and Computer Management** tools.
- Explore utilities like **Task Scheduler** and **Event Viewer**.
- Use **Resource Monitor** to analyze CPU, disk, memory, and network usage.
- Review logs and system processes and complete the lab questions.

TASK-1:



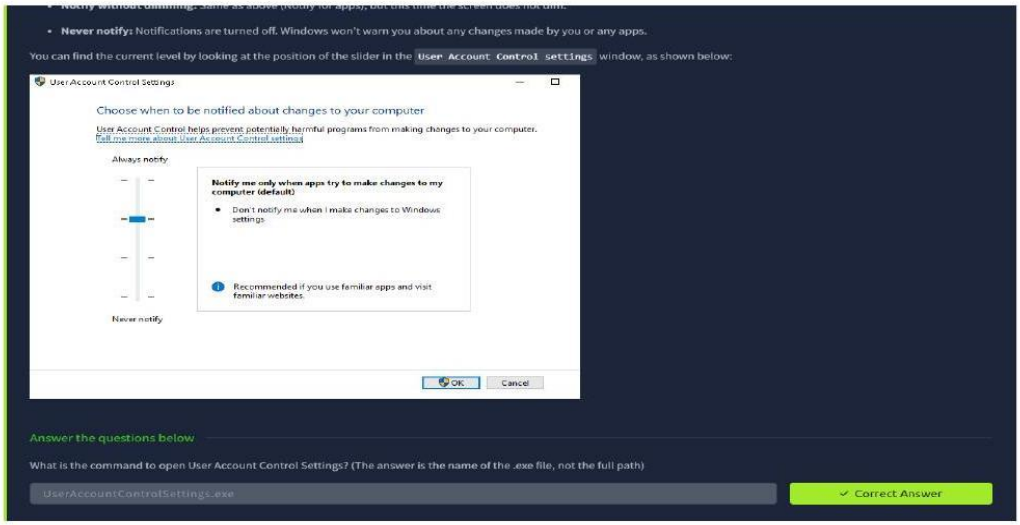
TASK-2:

TASK-2:



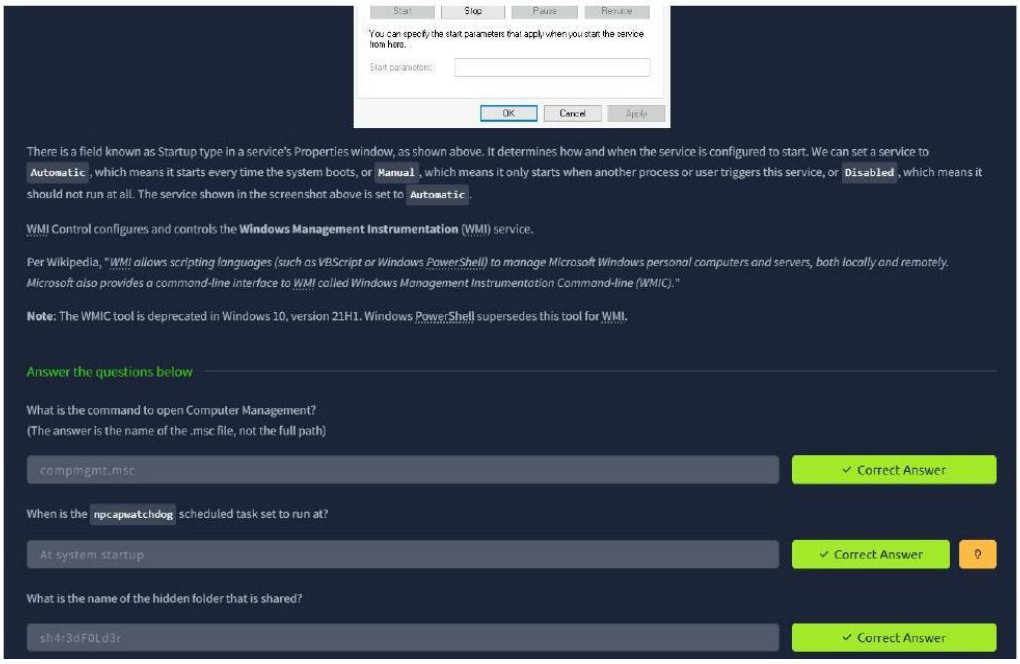
TASK-3:

TASK-3:



TASK-4:

TASK-4:



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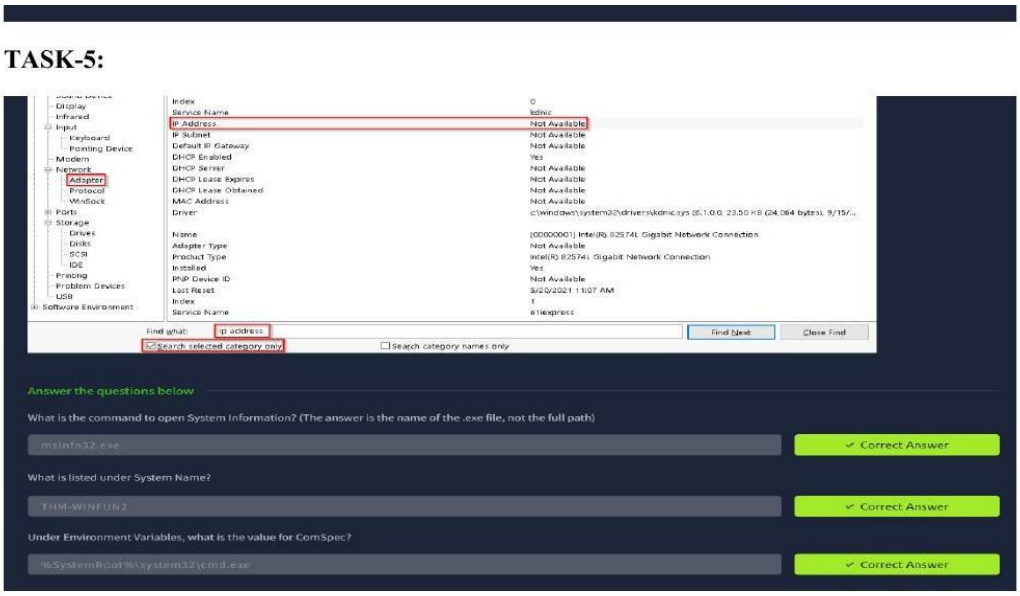

TASK-6:

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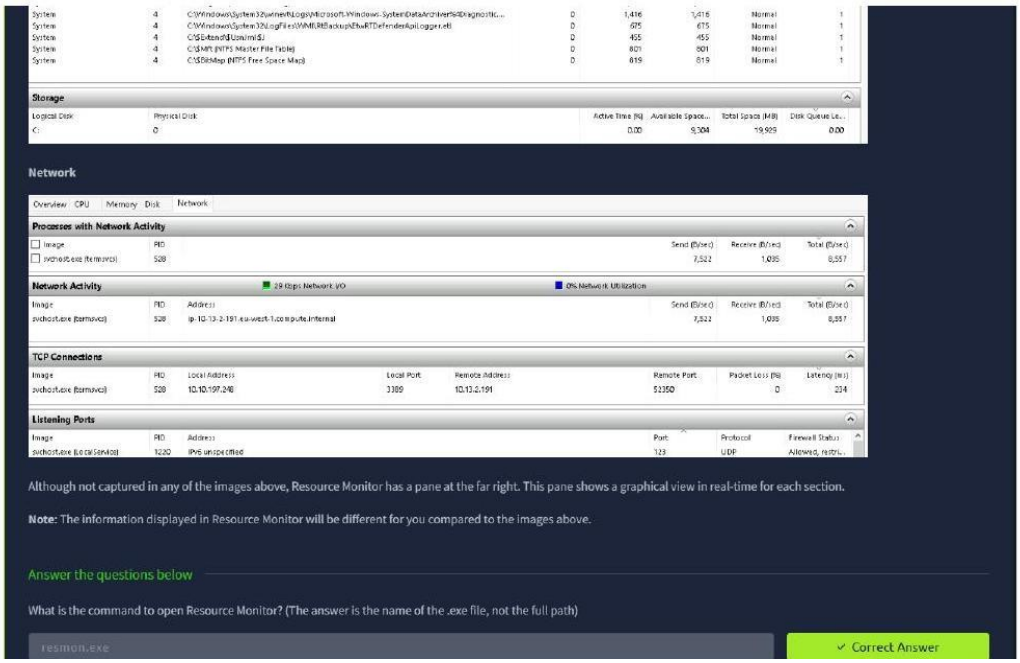
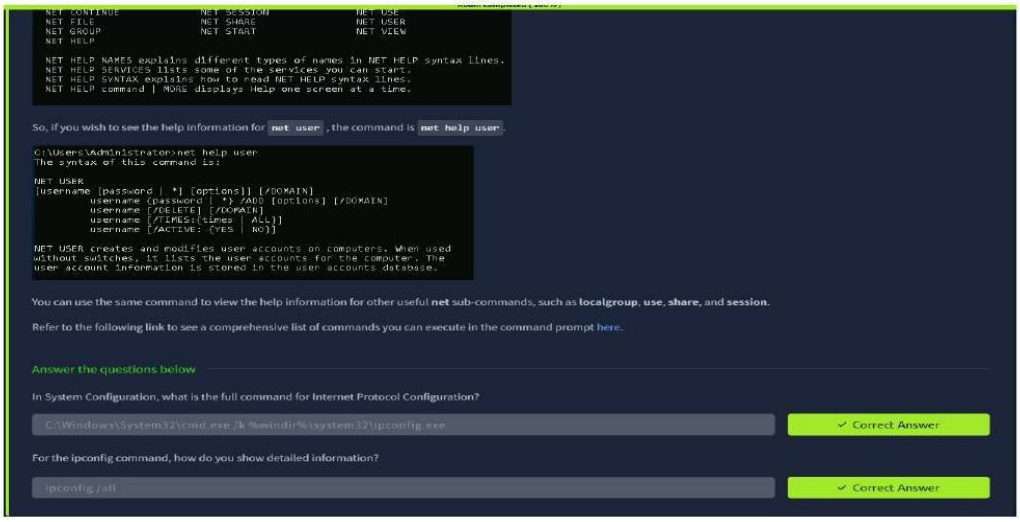


TASK-7:

TASK-7:



TASK-8:

TASK-8:

We're continuing with Tools that are available through the **System Configuration** panel.

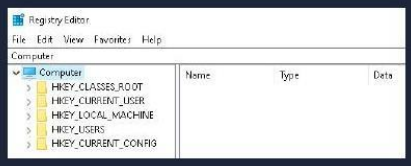
The **Windows Registry** (per Microsoft) is a central hierarchical database used to store information necessary to configure the system for one or more users, applications, and hardware devices.

The registry contains information that Windows continually references during operation, such as:

- Profiles for each user
- Applications installed on the computer and the types of documents that each can create
- Property sheet settings for folders and application icons
- What hardware exists on the system
- The ports that are being used.

Warning: The registry is for advanced computer users. Making changes to the registry can affect normal computer operations.

There are various ways to view/edit the registry. One way is to use the **Registry Editor** (`regedit`).



Refer to the following Microsoft documentation here to learn more about the Windows Registry.

Answer the questions below

What is the command to open the Registry Editor? (The answer is the name of the .exe file, not the full path)

regedit32.exe

✓ Correct Answer

?

TASK-9:

TASK-9:

Task 9 Conclusion

Recall that the tasks covered in this room were some of the tools that can launch from `MSConfig`.

Throughout the room, commands and shortcuts were shared for the utilities. This means you don't have to launch **MSConfig** to run these utilities.

You can also run some of these utilities directly from the Start Menu. See below where some of these utilities can be found.

Some of the tools listed in **MSConfig** that weren't mentioned in this room were either covered in Windows Fundamentals 1 or were left for you to explore on your own.

Answer the questions below

Read above.

No answer needed

✓ Correct Answer

RESULT:

Successfully learned the fundamentals of Windows through Task Manager, gaining hands-on experience with administrative tools, system configuration utilities, and the Windows Registry.

Windows Fundamentals-3

AIM:

To investigate and understand the **built-in security features of the Windows operating system**, encompassing Windows Update, Microsoft Defender Antivirus, Windows Defender Firewall, BitLocker Drive Encryption, and the Volume Shadow Copy Service (VSS) for system protection and recovery.

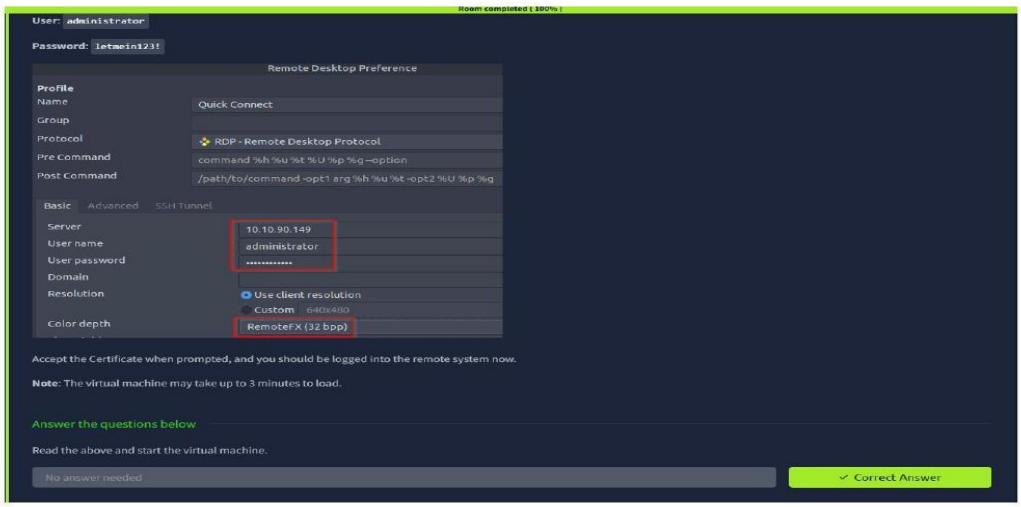
PROCEDURE:

- Start the virtual machine and log in via Remote Desktop Protocol.
- Open and navigate the Windows Security settings dashboard.
- Explore security features including Windows Update, Microsoft Defender, and Firewall settings.
- Examine BitLocker encryption configuration and system protection mechanisms.
- Study the Volume Shadow Copy Service (VSS) for backup and system restore functionality.

TASK-1:

- Open **Windows Security settings**.
- Explore security features such as **Windows Update**, **Microsoft Defender**, and **Firewall**.
- Examine **BitLocker encryption and system protection features**.
- Study **Volume Shadow Copy Service (VSS)** for backup and recovery.

TASK-1:

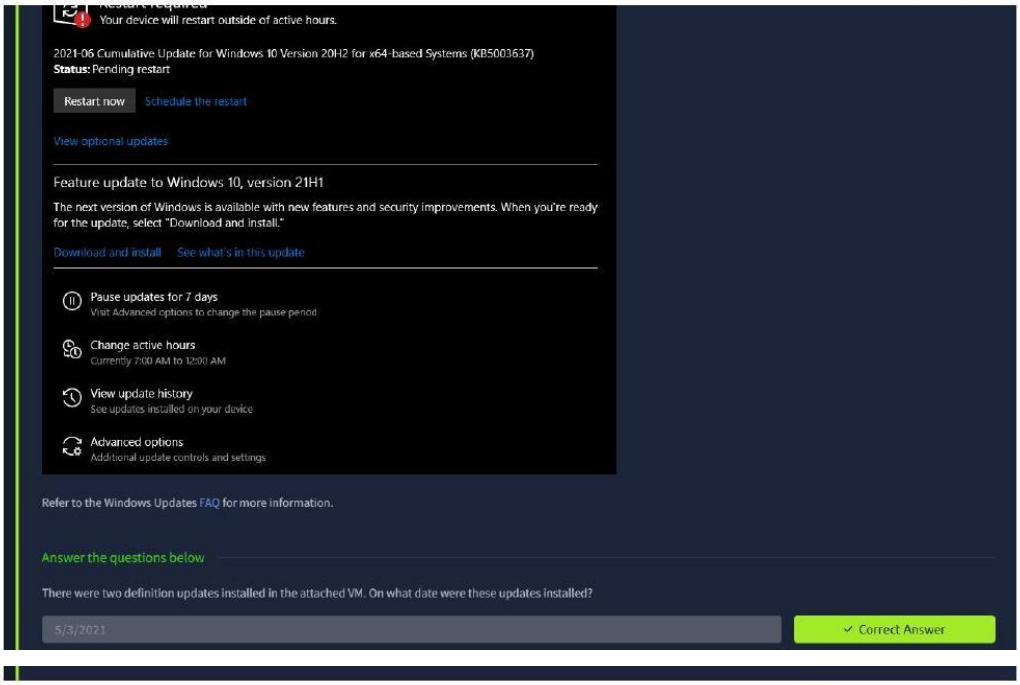


TASK-2:

TASK-3:

TASK-4:

TASK-2:



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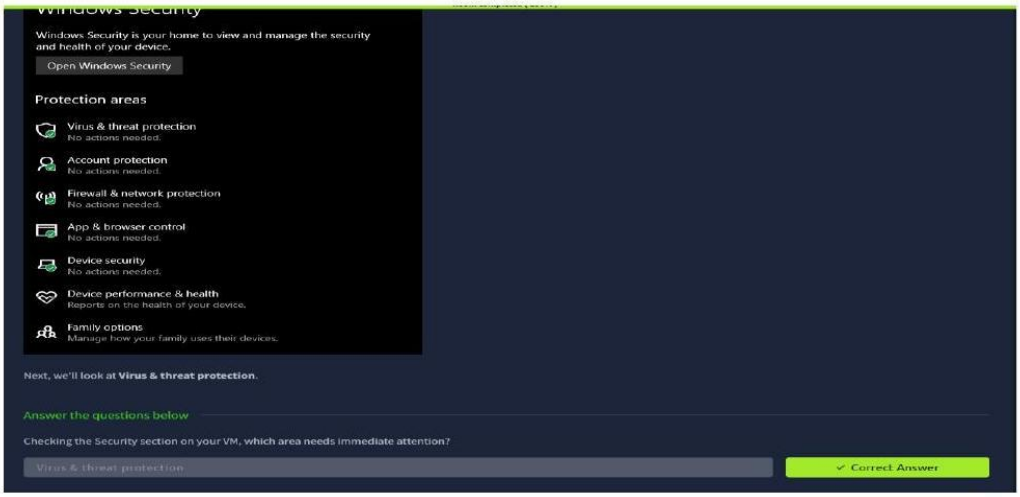
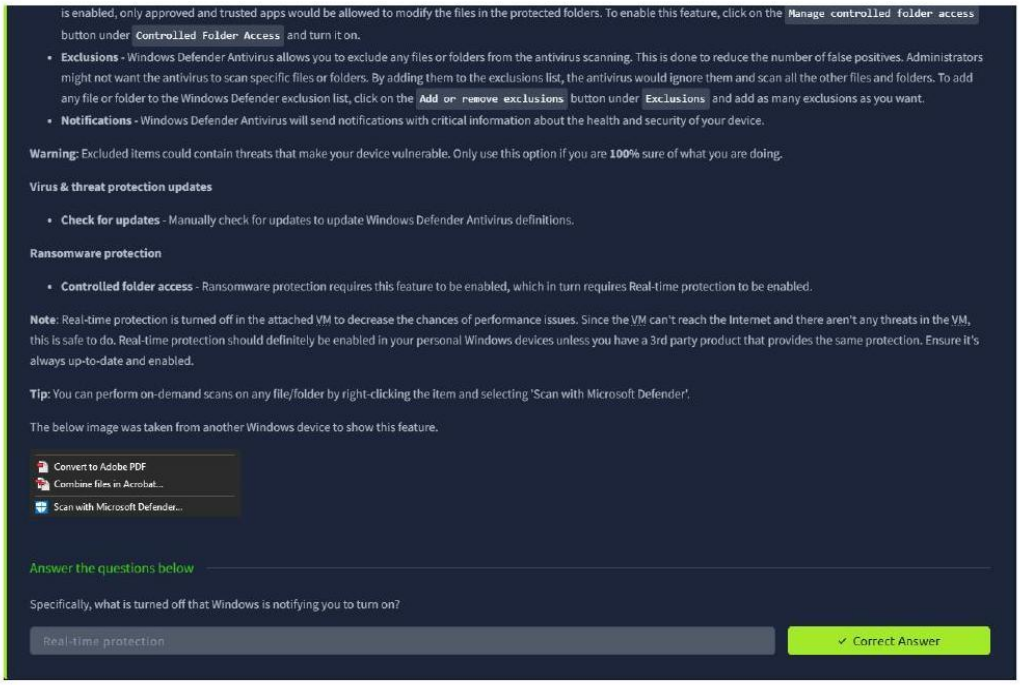


TASK-4:

TASK-5:



is enabled, only approved and trusted apps would be allowed to modify the files in the protected folders. To enable this feature, click on the `Manage controlled folder access` button under `Controlled Folder Access` and turn it on.

- **Exclusions -** Windows Defender Antivirus allows you to exclude any files or folders from the antivirus scanning. This is done to reduce the number of false positives. Administrators might not want the antivirus to scan specific files or folders. By adding them to the exclusions list, the antivirus would ignore them and scan all the other files and folders. To add any file or folder to the Windows Defender exclusion list, click on the `Add or remove exclusions` button under `Exclusions` and add as many exclusions as you want.
- **Notifications -** Windows Defender Antivirus will send notifications with critical information about the health and security of your device.

Warning: Excluded items could contain threats that make your device vulnerable. Only use this option if you are **100%** sure of what you are doing.

Virus & threat protection updates

- **Check for updates** - Manually check for updates to update Windows Defender Antivirus definitions.

Ransomware protection

- **Controlled folder access** - Ransomware protection requires this feature to be enabled, which in turn requires Real-time protection to be enabled.

Note: Real-time protection is turned off in the attached VM to decrease the chances of performance issues. Since the VM can't reach the Internet and there aren't any threats in the VM, this is safe to do. Real-time protection should definitely be enabled in your personal Windows devices unless you have a 3rd party product that provides the same protection. Ensure it's always up-to-date and enabled.

Tip: You can perform on-demand scans on any file/folder by right-clicking the item and selecting 'Scan with Microsoft Defender'.

The below image was taken from another Windows device to show this feature.

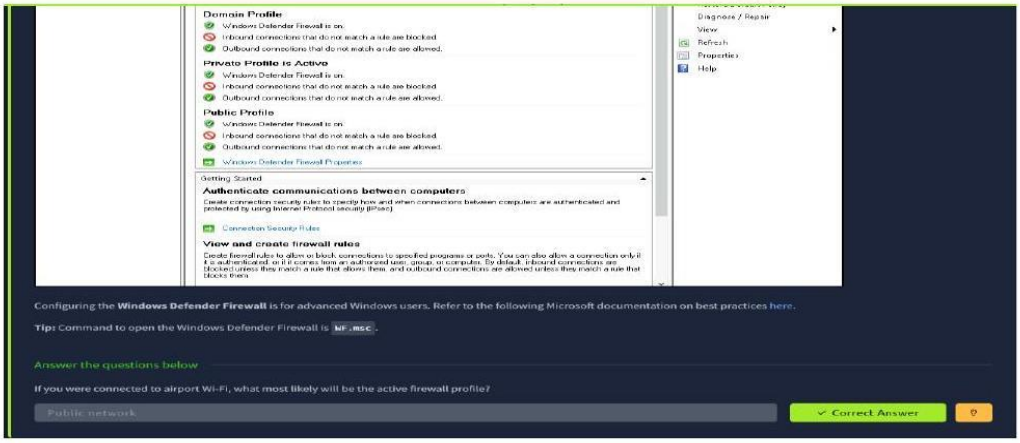
Answer the questions below

Specifically, what is turned off that Windows is notifying you to turn on?

Real-time protection

✓ Correct Answer

TASK-5:



Configuring the **Windows Defender Firewall** is for advanced Windows users. Refer to the following Microsoft documentation on best practices here.

Tip: Command to open the Windows Defender Firewall is `wf.msc`.

Answer the questions below

If you were connected to airport Wi-Fi, what most likely will be the active firewall profile?

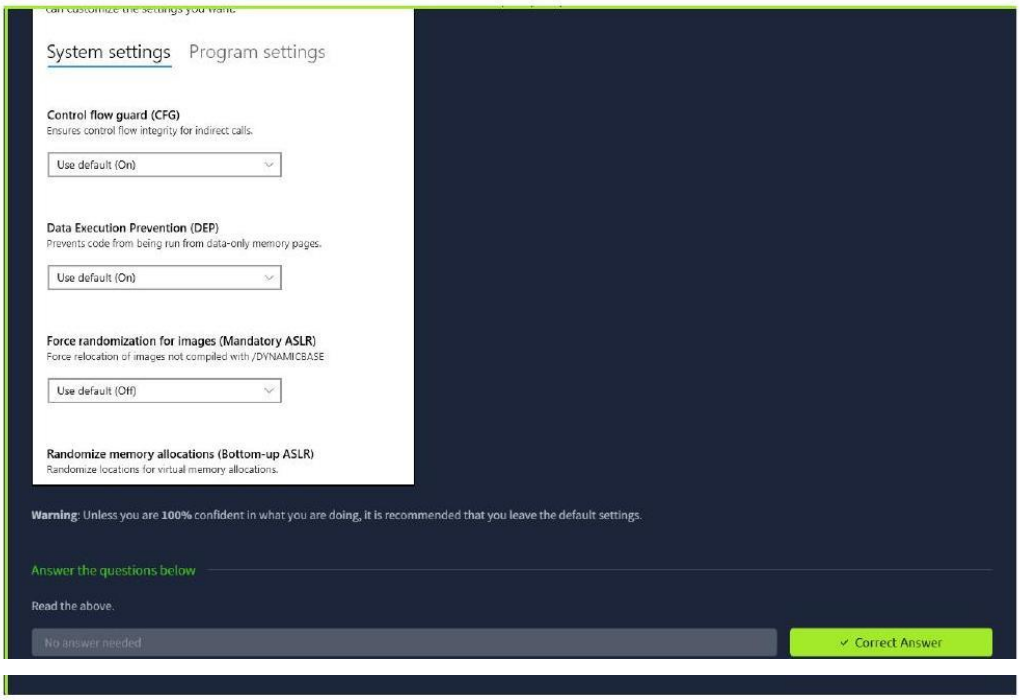
Public network

✓ Correct Answer

TASK-6:

TASK-7:

TASK-6:

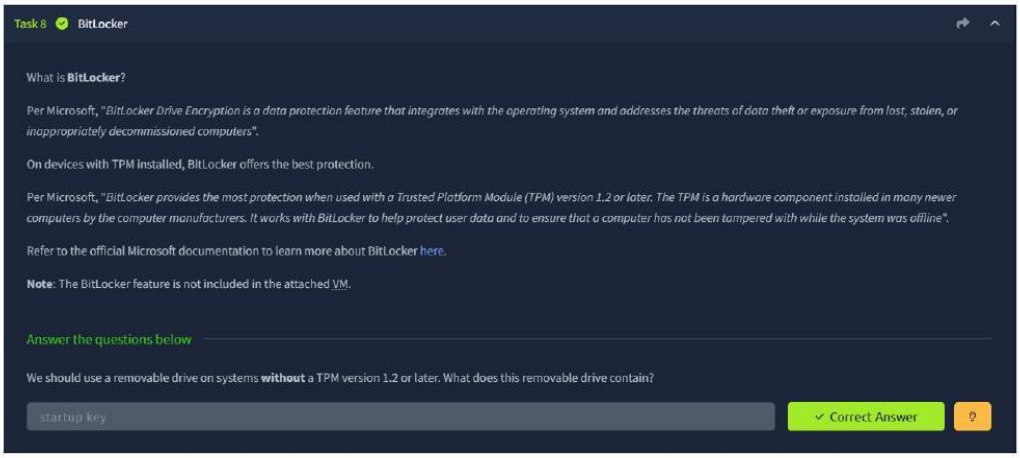


TASK-7:

TASK-8:

TASK-9:

TASK-8:



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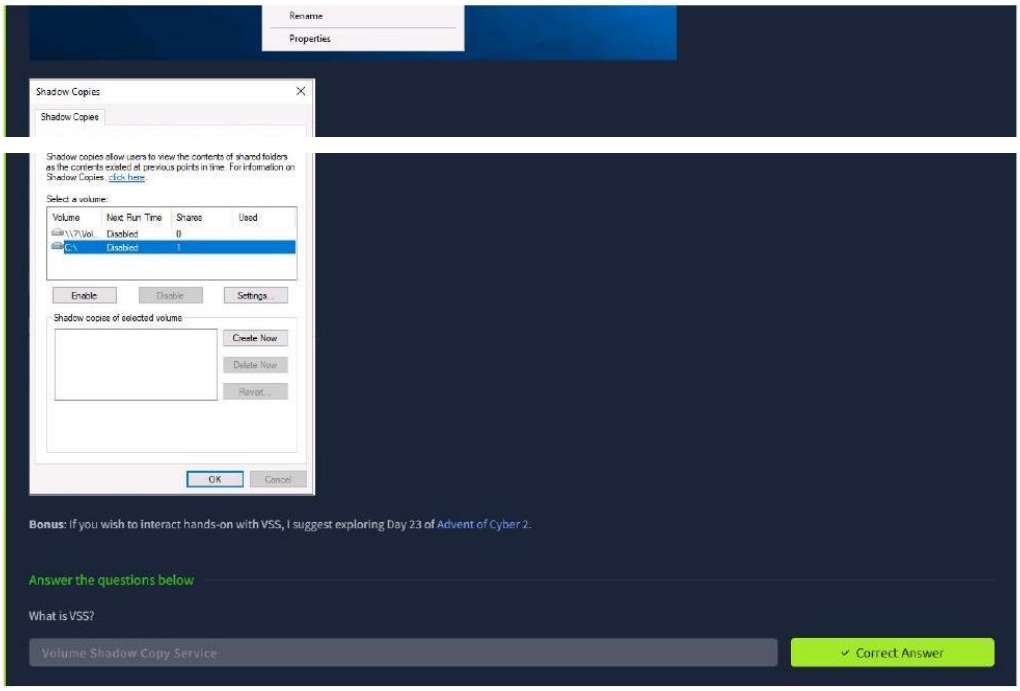

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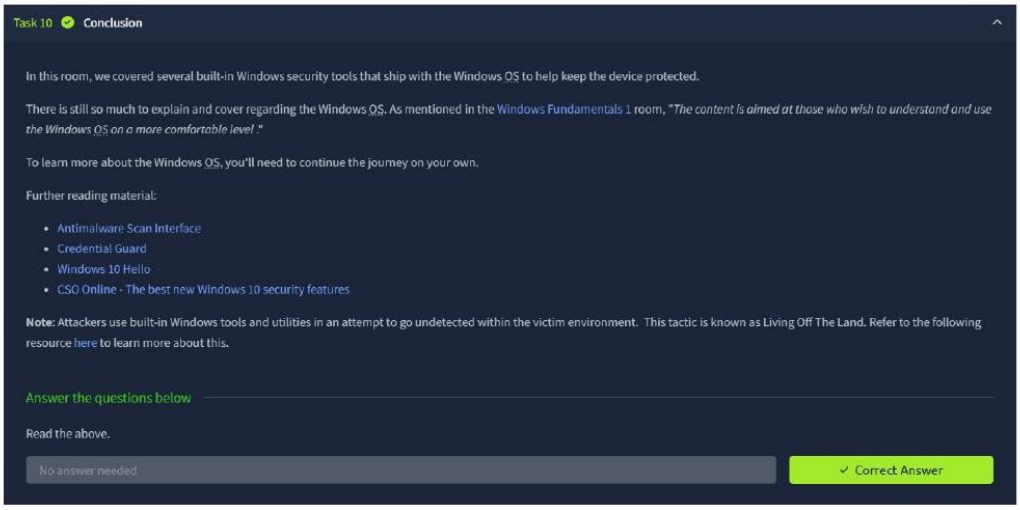
TASK-10:



Task 10 Conclusion

In this room, we covered several built-in Windows security tools that ship with the Windows OS to help keep the device protected.

There is still so much to explain and cover regarding the Windows OS. As mentioned in the Windows Fundamentals 1 room, "*The content is aimed at those who wish to understand and use the Windows OS on a more comfortable level.*"

To learn more about the Windows OS, you'll need to continue the journey on your own.

Further reading material:

- Antimalware Scan Interface
- Credential Guard
- Windows 10 Hello
- CSO Online - The best new Windows 10 security features

Note: Attackers use built-in Windows tools and utilities in an attempt to go undetected within the victim environment. This tactic is known as Living Off The Land. Refer to the following resource here to learn more about this.

Answer the questions below

Read the above.

No answer needed

✓ Correct Answer

RESULT:

All Windows Fundamentals modules were completed successfully. The built-in security features of the Windows operating system, including threat protection, firewall management, encryption tools, and recovery mechanisms, were thoroughly explored.